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ORIGINAL PAPER

Experimental evidence in a poison frog model suggests that tadpole transport on the dorsum may affects warning signal effectiveness in poison frogs

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Abstract
Aposematism is an anti-predator strategy where predators learn to associate the warning signal on prey with an unpleasant experience, and consequently, avoid attacking similar prey in the future. Conspicuous coloration in poison frogs (Dendrobatidae) is considered a warning signal. During parental care, parents transport their tadpoles on the dorsum, which could alter the detectability and recognition of such warning coloration by visually oriented predators. We tested this hypothesis using domestic chicks trained to avoid and discriminate between printed frog models with and without conspicuous-warning coloration. We tested whether the chicks recognized the warning coloration on printed frog models that varied in the quantity of tadpoles on the dorsum. Chicks first attacked frog models without warning coloration, whether they had tadpoles on the dorsum or not. In contrast, frog models with warning coloration were attacked last by chicks. Moreover, the frog models with warning coloration and without tadpoles experienced a lower risk of attack by chicks than similar frog models with tadpoles. However, aposematic frog models maintained the warning function of conspicuous coloration if it was located on parts of the parent's body that are not covered by the tadpoles when transported. Our results suggest that tadpoles on the dorsum of parents might compromise the effectiveness of the warning signal display in poison frogs increasing the risk of attack by visually oriented predators.

Keywords Aposematism · Parental care · Natural selection · Dendrobatidae · Tadpole transport

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Introduction

Predation promotes the evolution of adaptations that increase the probability of survival in prey organisms (Caro 2005; Ruxton et al 2018). One defensive adaptation is crypsis which is when organisms exhibit phenotypic characteristics that make them similar to their environment or break up their body outline, hindering detection by a predator (Cott 1940; Stevens and Merilaita 2011; Honma et al 2015; Nokelainen and Stevens 2016). Another defensive adaptation is aposematism, which consists of exhibiting generally conspicuous signals (e.g., bright coloration) to inform and warn potential predators about a noxious or unpleasant secondary defense (e.g., toxins or unpalatability) (Poulton 1890; Cott 1940; Ruxton et al 2018). Theoretically, for aposematism to become established and functional, predators should learn to associate a warning signal in a prey organism with a negative or unpleasant experience; and after that, avoid attacking individuals with a similar phenotype in future encounters (Endler 1991; Endler and Mappes 2004; Lynn 2005). Therefore, for such learning to happen, the warning signal in the prey should be easily detectable and recognizable by the predator (Guilford 1986; Sherratt and Beatty 2003). Aposematism has evolved multiple times in plant and animal lineages and there is evidence in favor of its role as an effective defensive adaptation (Dumbacher et al 1992; Lev-Yadun 2003; Caro 2005; Stevens and Ruxton 2012; Brandley et al 2016; Rojas 2016; Banci et al 2020). However, the origin of aposematism and its variability in animal lineages remains controversial and is a topic of great interest in evolutionary biology (Mappes et al 2005; Briolat et al 2019).

The family Dendrobatidae (poison frogs) consists of approximately 200 species distributed in the Neotropical region (Frost 2022). Some species of this family are considered aposematic due to the presence of conspicuous coloration and toxins in their body (Daly 1982; Santos and Cannatella 2011; Kahn et al 2016; Rojas 2016). Both comparative phylogenetic analyses and predation experiments with printed models that emulate poison frogs (using chicks as model predators in diverse scenarios) have been commonly used to investigate the origins of aposematism in vertebrates (Santos et al 2003; Darst et al 2006; Saporito et al 2007; Wang and Shaffer 2008; Tarvin et al 2017). In addition, poison frogs have been a prominent system model for understanding the relationship between aposematic defense adaptation and diet (Caldwell 1996; Santos et al 2003), metabolic rate (Pough and Taigen 1990; Santos 2012), diversification of acoustic communication signals, speciation rates (Amézquita et al 2011; Santos et al 2014), agonistic interactions between conspecifics (Pröhl 2005; Crothers et al 2011), and the evolution of parental care (Weygoldt 1987; Carvajal-Castro et al 2021). Other interesting aspects related to aposematism in poison frogs are understudied, for example, sexual dichromatism, the role of conspicuous coloration in intrasexual competition, and the interaction between parental care and components of warning coloration signaling (Rojas et al 2018; Palacios-Rodríguez et al 2022; but see Seshadri and Thaker 2022).

Poison frogs exhibit different levels of complexity in parental care depending on the species (Weygoldt 1987; Brown 2013; Summers and Tumulty 2014). Generally, parental care includes egg attendance in terrestrial microhabitats (i.e., leaf litter) and posterior tadpole transport on the dorsum of parents to diverse types of bodies of water (e.g., ponds, small streams, and phytotelmata) where they complete their development. In some specialized cases, tadpoles are fed by the mother with unfertilized eggs (Weygoldt 1987; Wells 2007; Summers and Tumulty 2014). Among poison frogs, the average number of tadpoles simultaneously transported on the dorsum of a parent varies between 1 and 33 in aposematic species and between 1 and 25 in species traditionally considered non-aposematic

(Carvajal-Castro et al 2021). How much time those tadpoles spend on the dorsum of their parents is unknown for most species (Wells 2007). However, it can last hours or days in some species (Pröhl and Hödl 1999; Killius and Dugas 2014; Beck et al 2017; Pašukonis et al 2019), while in others, tadpoles may remain on the dorsum of the parents for several days (e.g., *Colostethus inguinalis* Cope 1868), and even grow while being transported (Wells 1980).

Parental care enhances offspring fitness but often is costly for parents in terms of a reduction in the future survival and reproduction chances (Clutton-Brock 1991; Royle et al. 2012). In poison frogs, during tadpole transport on the dorsum, the carrier incurs costs associated with increased energy expenditure, physiological wear, potential loss of territory, and a reduction in mating opportunities (Wells and Taigen 1992; Navas 1996; Downie et al 2005; Bell 1990; Beck et al. 2017). Another potential but untested cost of tadpole transport in poison frogs may be an increased risk of predator attacks. If tadpoles on the dorsum cover or alter the detectability and recognizability of the warning signal coloration of the parent, both the parent and the offspring could be susceptible to attacks by visually oriented predators. Under this scenario, it can be expected that natural selection might contribute to shaping the distribution of color patterns on the bodies of aposematic poison frogs to minimize the cost associated with tadpole transport. For example, warning coloration could be distributed towards the anterior part of the body and/or limbs (tadpoles are carried in the mid to upper back). In that way, the transmission of information to potential predators would not be compromised during the transport of tadpoles on the dorsum.

In addition to the color body distribution mentioned above, warning coloration in poison frogs could be easily detected and recognized by predators if, during parental care, adults transport a few tadpoles simultaneously. However, factors like distance between parent territory and bodies of water, characteristics of the site where the tadpole develops, and size of tadpoles, may have influenced the quantity of offspring transported simultaneously on the dorsum by parents (Aichinger 1991; Wells 2007; Ringler et al 2013, 2016; Pašukonis et al 2017, 2019). In fact, a reduction in the number of eggs per clutch, a trait positively correlated with the number of tadpoles transported simultaneously on the dorsum, happens in aposematic species that deposit their tadpoles on arboreal phytotelmata (Lehtinen et al 2004; Summers and McKeon 2004; Summers and Tumulty 2014; Carvajal-Castro et al 2021). This trend may be due to the energetic constraints of the parents and limitations in space and nutrients in phytotelmata (Summers and McKeon 2004; Carvajal-Castro et al 2021). Therefore, the number of tadpoles transported simultaneously on the dorsum of the parents in poison frogs would not necessarily be a life history trait mainly shaped by predation risks but by ecological *trade-offs* in different contexts.

In this study, we tested for potential costs associated with parental care in poison frogs, and their relationship with conspicuous color patterns. We experimentally tested whether tadpole transport on the dorsum of parents in poison frogs affects how recognizable a conspicuous (warning) coloration pattern is by learned visual-oriented predators. Also, we tested whether such recognition by predators changes with the distribution of conspicuous color on the body of the frog. To do so, we used printed frog models emulating aposematic and non-aposematic poison frogs with different quantities of tadpoles (model tadpoles) on the dorsum (Fig. 1) and compared the order and latency to attack upon them by a trained predator, the domestic chick *Gallus gallus domesticus*. Specifically, we tested the following three predictions (Fig. 2): First, frog models with warning coloration and model tadpoles on the dorsum will have a higher risk of attack by chicks than frog models with a lower number (or absence) of model tadpoles on the dorsum. Second, frog models (both with and without model tadpoles) with warning coloration on parts of the body that would not

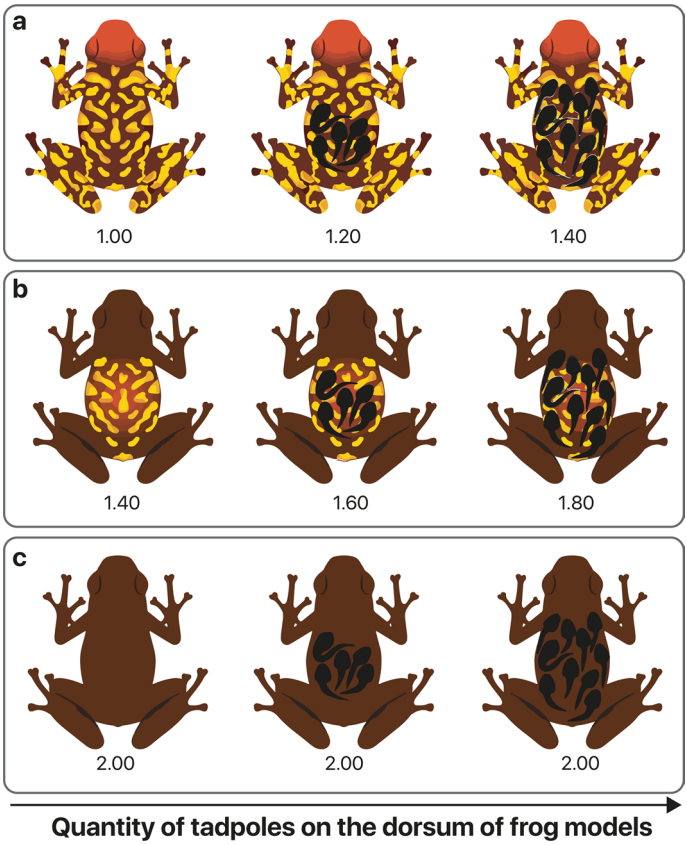

Fig. 1 Representation of the three types of frog models used in the experiments of attacks by trained chicks. **a** Frog models emulating a poison frog with full warning coloration in the body. **b** Frog models with warning coloration restricted to parts of the body potentially covered by tadpoles during transport on the dorsum of the parental individual. **c** Frog models emulating a non-aposematic species (without warning coloration). The difference among frog models in the same lies in the quantity of tadpoles on the dorsum, and their potential effect on the recognition of the warning signal (coloration) by a visually oriented predator (domestic chick *Gallus gallus domesticus*). The numbers below each frog model refer to the score value given to each attack made by the chick according to the frog model (see details in methodology text, Phase III).

be covered by model tadpoles (i.e., head and limbs) will have a lower risk of attack by chicks than frog models without warning coloration or than frog models with warning coloration on parts of the body that the presence of model tadpoles would cover. Third, frog models without warning coloration and with or without the presence of model tadpoles on the dorsum will not differ in the risk of attack by chicks. Evidence to support these three predictions in laboratory experiments would highlight a previously untested cost of parental care in poison frogs and suggest that predation risk during the tadpole transport phase may have contributed to shaping the evolution of warning coloration in poison frogs. Since the parental behavior of tadpole transport on the dorsum is thought to have evolved before aposematism in poison frogs (Carvajal-Castro et al 2021), it is expected that selection has favored the fitness of those individuals and species with warning coloration on parts of the body that are visible despite the presence of offspring on the dorsum. This study includes

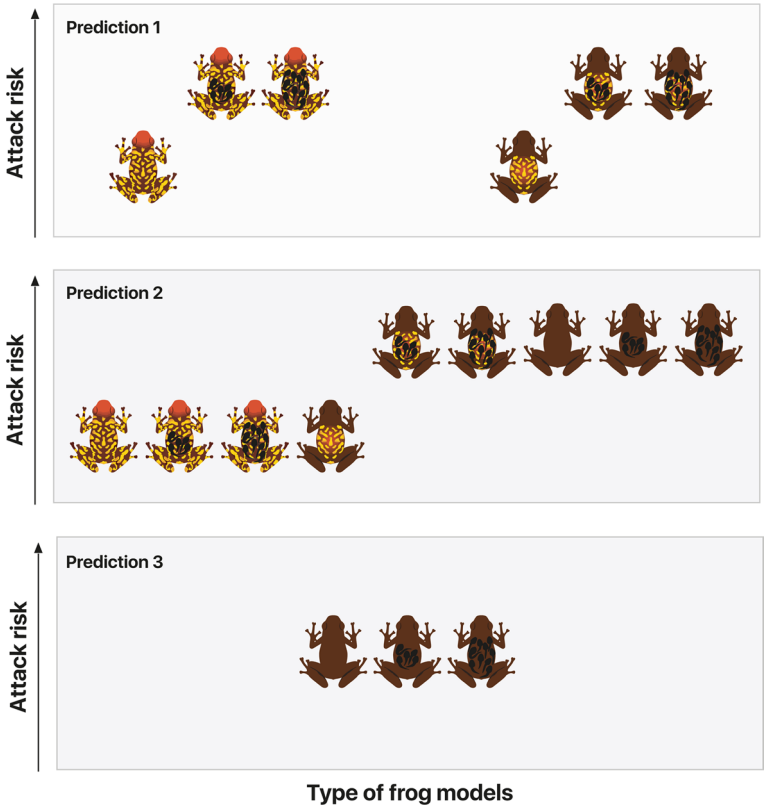

Fig. 2 Diagram showing three-predictions in this study. In general, (1) frog models emulating aposematic species with warning coloration and the presence of tadpoles on the dorsum will have a higher risk of attack by trained chicks than frog models without tadpoles on the dorsum, (2) frog models emulating species with warning coloration on parts of the body that transported tadpoles would cover will have a lower risk of attack by trained chicks than frog models emulating species without warning coloration or with warning coloration on parts of the body the presence of tadpoles would cover, and (3) frog models emulating non-aposematic species without warning coloration will have a similar risk of attack by trained chicks whether they have tadpoles on the dorsum or not

a novel perspective that would help elucidate the cost of parental care and its potential importance in processes underlying the evolution of warning signals in poison frogs and aposematic animals in general.

Materials and methods

Experimental models design

The experimental frog models (Fig. 1) emulated the coloration pattern, body shape, and size of adult individuals of a morph of the *Oophaga histrionica* complex (Lötters et al 2007; Frost 2022). This morph inhabits tropical rainforests in the Pacific region of Colombia and was selected as a reference for designing frog models because its warning signal includes red and yellow colors, which are present in many other aposematic species

(Svádová et al 2009; Amézquita et al 2013; Ruxton et al 2018). In this study, we did not want to replicate specific situations for a species *per se* (i.e., number and size of tadpoles transported), but rather, how the presence of tadpoles with non-aposematic coloration (McDiarmid and Altig 1999) would hinder the transmission of warning signals in poison frogs.

We designed three types of frog models for the experiments. The first type of frog model emulated a species with warning coloration throughout the body. This type of frog model allows an assessment for the effectiveness (decreased risk of attack) of coloration when it is present on the body parts that are not covered by the tadpoles when transported (Fig. 1a). The second type of frog model represented a hypothetical species with warning coloration restricted to the dorsum of the individual, which is where the tadpoles would be placed during transport to bodies of water. With this frog model, we assessed whether transported tadpoles interfere with the effectiveness of conspicuous coloration as a warning signal to a potential predator (Fig. 1b). The third type of frog model represented a non-aposematic species (i.e., without warning coloration; Fig. 1c). Furthermore, the three types of frog models were modified to emulate situations of no tadpoles on the dorsum and different quantities of tadpoles (five and ten tadpoles), covering approximately 10 and 20% of the total area in the frog model. We considered the number of tadpoles used in our frog models as representative of the variability in this natural history trait for poison frogs (Carvajal-Castro et al 2021).

The colors used in the three types of frog models came from calibrated photographs (i.e., white balance and chromatic levels adjusted with a Kodak color card Q13, CAT 152 7654) of a morph of *O. histrionica* complex (aposematic species) and the dorsal part of *Leucostethus brachistriatus* (Rivero and Serna 1986) (non-aposematic species). Given that in nature, tadpoles of poison frogs tend to be dark-colored (McDiarmid and Altig 1999; Lötters et al 2007), tadpoles in our experimental frog models were printed black. Frog models had a snout-to-vent length of 50 mm and were printed in a circular cream-colored background (C:2.49, M:12.15, Y:27.35, K: 0.2) of radius = 4 cm. All the frog models were produced using Adobe Illustrator version 13.0.1 and printed on 250 g Kodak matte paper with a Xerox 550 laser printer (Norwalk, Connecticut, USA).

Experimental predator and housing

To assess the potential costs of transporting tadpoles on the dorsum in poison frogs, we used domestic chicks *Gallus gallus domesticus*. Domestic chicks as a proxy of natural avian predators have been used in studies about the ecology and evolution of aposematism in diverse organisms (Schuler and Hesse 1985; Roper and Wistow 1986; Roper and Cook 1989; Schuler and Roper 1992; Gamberale-Stille and Tullberg 1999, 2001; Hauglund et al 2006; Lawrence and Noonan 2018). Moreover, avian vision is well documented at the physiological level (Osorio et al 1999; Matsushima et al 2003), and it is known that chicks can learn to discriminate and categorize different colors and patterns (Ham and Osorio 2007; Aronsson and Gamberale-Stille 2008).

For our study, we used 84 newly hatched (i.e., two days old) male chicks from a commercial hatchery. The chicks were individually marked and housed in 1 m^2 wooden cages. As the chicks were young, they were heated with 25 W bulbs and fed *ad libitum* with water and corn crumbs. For the food motivation of the chicks during the experiments, we restricted food for 2 h before each experimental session. After experiments, chicks were

donated to an agroecological farm (See ethics declarations for permits and animal treatment procedures).

The experimental arena and chicks pre-experimental training

Pre-experimental training and experiment took place in a rectangular arena (100 cm long, 60 cm depth, 50 cm height) made of Triplex wood (Fig. 3, Video S1); this arena included an expandable experimental zone and a companion zone. This companion zone consisted of a small area where focal chicks habituated to the arena, and two additional chicks (companion chicks) were present during experiments (see below); the presence of these companion chicks helped to reduce stress in the focal chicks during the study (Dixon et al 2014). Chicks were trained to attack and feed on dry mealworms *Tenebrio molitor*, which were placed in Petri dishes distributed on the ground of the experimental zone. Once the chick consumed all the mealworms, the pre-training was over, and they moved on to the next Phase of the study.

Phase I: learning and discrimination of the warning signal

Chicks can learn to discriminate between frog models that imply or not an unpleasant experience (Roper and Wistow 1986; Skelhorn et al 2016). Such learning in chicks was achieved by using two groups of mealworms as stimuli. The first group consisted of unpalatable mealworms soaked in a 10% chloroquine phosphate solution, an alkaloid with an unpleasant taste but not toxic to birds (Lindström et al 1999). These mealworms were placed upon frog models emulating aposematic species (i.e., with full warning-conspicuous coloration). The second group consisted of palatable mealworms (untreated in chloroquine phosphate solution). Palatable mealworms were placed upon frog models emulating non-aposematic species (i.e., without warning-conspicuous coloration).

Phase I was performed using frog models with warning coloration on the entire body (i.e., full warning coloration) and without warning coloration, both without tadpoles on the dorsum (Fig. 1a and c) and their corresponding mealworm. This phase consisted of seven learning trials, one per day for one week. Before each session, the focal chick was

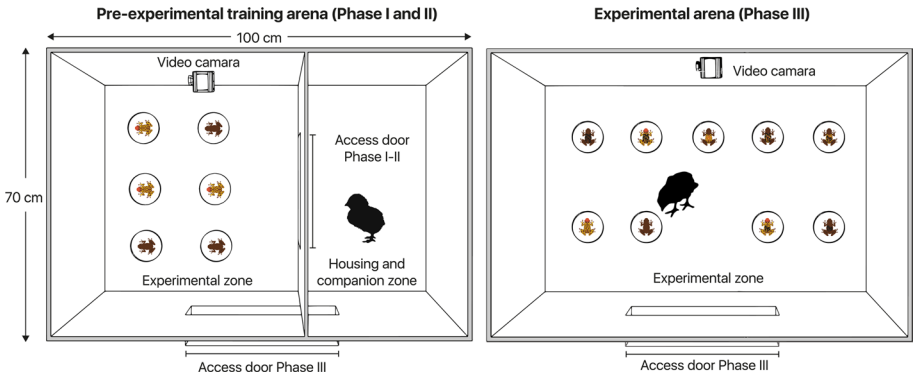

Fig. 3 Top view of the arena used for experiments. The access doors refer to the initial location of the predator (chick *Gallus gallus domesticus*) during each Phase of this study (see methodology for details). Left panel: Phases I and II (learning, avoidance, and discrimination). Right panel: Phase III (warning coloration recognition with different levels of visual interference).

placed for 30 min in the companion zone of the experimental arena to become habituated. Each session began by placing the focal chick in front of a door that divided the experimental zone from the companion zone (Fig. 3). Once the chick crossed the door, it was simultaneously presented with six frog models (3 frog models with full warning coloration and 3 frog models without warning coloration) randomly distributed in the experimental zone. The time allowed for each chick to attack (i.e., peck, taste, and/or eat) the mealworms placed on the frog models was five minutes (i.e., 300 s) in this and the other two phases. This time range was selected based on previous studies using similar experiments (Lawrence and Noonan 2018; Lawrence et al 2019). Focal chicks were allowed to view two companion chicks located in the companion zone (Dixon et al 2014). Companion chicks were not included in the statistical analyses. We also excluded focal chicks that did not show any interest in feeding (e.g., they refused to attack any mealworm, fell asleep, and tried to escape from the experimental arena) in three or more trials of this Phase (Lawrence et al 2019).

We used an electivity score index (ESI) to quantify the chicks' learning throughout Phase I. This index assigns scores according to the type of frog model attacked by the chick (Amézquita et al 2013). If the chick attacked an unpalatable mealworm upon a frog model with warning coloration, it was scored = 1 (incorrect decision); conversely, if the chick attacked a palatable mealworm upon a frog model without warning coloration, it was scored = 2 (correct decision). The ESI weights scores according to the attack order by the chick; the score of the first attack was multiplied by a value of 1, and the score of the second attack was multiplied by a value of 0.9, and so on until reaching a value of 0.1 (Amézquita et al 2013). In addition, the ESI considers repeat attacks, i.e., when a chick attacked a given mealworm a second time after attacking other mealworms; the score for these repeat attacks was half the original score. Thus, the ESI incorporated the encounter rate required by a predator to assess a given prey type (Sherratt 2011; Hämäläinen and Thorogood 2020).

The sum of the scores obtained (i.e., attack score * order weighting) was divided by the total number of attacks made by the chick during the learning session. Hence, chicks that performed more accurate discrimination between models with or without warning coloration (i.e., made better attack decisions) obtained a higher ESI score than chicks that did not discriminate accurately between frog models. The mathematical expression of the ESI is:

$$\mathrm{ESI} = \frac{\sum \text{(prey score)(weight by attack order)}}{\text{total number of attacks made by the chick in the learning session}}$$

Phase II: learning and avoidance of the warning signal

Chicks might exhibit individual differences in learning ability (Rowland et al 2017; Skelhorn et al 2016). Therefore, after Phase I, we conducted a test to corroborate the avoidance of warning coloration in an aposematic frog model by each chick. In this test, we presented, individually and sequentially, a frog model with full warning coloration (without tadpoles) and its respective unpalatable mealworm to each chick. The presentation of each frog model to the chick lasted five minutes, followed by five minutes resting period. If the chick attacked the mealworm upon the frog model, the sequence continued up to a maximum of ten presentations of the frog model. When the chick refused to attack the mealworm upon the frog model for three consecutive presentations, the chick was considered to

have learned to avoid the prey through the association between the warning coloration and the unpleasant taste of the mealworm (Exnerová et al 2015; Lawrence et al 2019).

Once the chick avoided attacking a mealworm upon the frog model, we offered a palatable mealworm upon a frog model without warning coloration. This attack allowed us to discard the possibility that a chick would not attack previously presented mealworms due to satiation. If chicks went through ten presentations of frog models with full warning coloration without three consecutive refusals, they were not considered to have learned the association between the warning coloration and the unpalatable mealworm and were not included in the statistical analyses (Lawrence et al 2019). Chicks that completed this session moved on to Phase III.

Experiment

Phase III: recognition of the warning signal in frog models with and without the presence of tadpoles

To assess the potential predation risk that a parental poison frog would assume when transporting its tadpoles on its dorsum, we conducted an experiment with the trained chicks. This experiment consisted of a session in which each chick had available all frog models used in this study (Fig. 1). During this Phase III, we used only palatable mealworms; in this way, we could attribute the attacks by the chicks to their perception of the frog model rather than to characteristics of the mealworm *per se* (e.g., shape, odor, consistency). The placement of the frog models in the arena was randomly established (using the function "random.sample" in R software) before presenting the focal chick (Fig. 3). The chick was removed from the arena after it attacked all available frog models or after 5 min (Jöels et al 2006).

The behavior of each chick during Phase III was classified according to its attack preferences for the frog models presented. To do so, each frog model had a score value assigned according to the type of frog model and the presence of model tadpoles. As with the ESI in Phase I, a correct attack decision by the chick was scored as = 2 (frog model without warning coloration), whereas an incorrect attack decision was = 1 (frog model with full warning coloration). In general, the score value increased from the frog model with full warning coloration without model tadpoles on the dorsum to the frog model without warning coloration. In each type of the frog model, the score values increased by a factor of 0.20 from the frog model without tadpoles to the frog model with many tadpoles. In this way, the score obtained by a chick also reflected the effect of tadpoles on the dorsum of frogs in the perception of the warning signal displayed by the chicks. There are two exceptions to this 0.20 points increasing rule. First, the same score (1.4) was assigned to two frog models because they are assumed to have a similar exposed area of conspicuous coloration (see Fig. 1). Second, frog models without warning coloration had an equal score value (2.0) because, in this case, we assumed that whether or not tadpoles were present, the frog model maintained its non-conspicuous coloration (Fig. 1c) and attacking any of them would be a correct attack decision by the chick.

The attack behavior of the chicks was represented by the sum of the score values associated with each frog model attacked. For this sum, we considered only those values corresponding to the first three chick attacks because, from the nine available options, only three frog models would correspond to non-aposematic species. That is, if the chick recognized

and performed correct discrimination and avoidance of the warning coloration in the frog models, the attacks would be of the mealworms on the three frog models without warning coloration.

Data analysis

All trials performed in the three Phases were videotaped and subsequently analyzed in BORIS v.7.7.3 (Friard and Gamba 2016). All videos were analyzed by the same person (first author) to avoid potential biases associated with variability in the measurement criteria. We recorded the type and order of the frog model attacked, the total number of attacks performed by the chick during each session, the latency time for attacking frog models, the state of the frog model (i.e., attacked or avoided), and the duration of the session. Below we describe the statistical analyzes performed in Phase I and III. In Phase II, there were no specific statistical analyses.

Phase I: learning and discrimination of the warning signal

We calculated the electivity score index (ESI) for each chick in the learning and discrimination trials. To test whether the chicks learned to discriminate between frog models with or without warning coloration, a repeated-measures ANOVA was performed. In this analysis, the explanatory variable was the number (seven) of learning trials, and the response variable was the ESI score. In addition, a paired t-test was used to corroborate that the ESI score in the last session of the learning and discrimination test (i.e., session seven) was significantly higher than the ESI score in the first session of this test.

Phase III: recognition of the warning signal in frog models with and without the presence of tadpoles

We ran a non-parametric Friedman test to assess whether chick attack order was associated with a type of frog model and with the presence or absence of model tadpoles. In this analysis, the response variable was the chick attack order, and the explanatory variable was the frog model type. Given that we used nine frog models in the experiment associated with this test, the attack order should vary between 1 and 9; however, frog models that were not attacked by chicks were included in the analysis as attack order = 10. In this way, we differentiated those frog models that were not attacked from those that were attacked last. Moreover, the Friedman test included the identity of the chick to account for attacks made by the same individual. A post hoc Nemenyi test was used to evaluate pairwise differences in chick attack order between different color pattern/tadpole quantity combinations (Siegel 1956; Lindström et al 1999; Hatle and Salazar 2001).

To assess whether the attack behavior of the chicks toward the mealworms on the nine frog models (Phase III) was random or not, a chi-square test (χ^2) was performed. In this test, we used the chick attack score, i.e., the sum of the score for the first three attacks on the frog models. Since we have nine frog models, the possible resulting chick attack scores are limited; in our analysis, we considered only the first three attacks made by the chicks, so there are only 13 possible chicks attack scores. We used those 13 chicks attack scores to calculate both the observed and expected frequency distribution. Given that the expected frequency was < 5 (32 chicks / 13 categories of chick attack scores = 2.46) we obtained, a p-value relative to a null distribution of χ^2 values, generated through a 10,000-sample

computational simulation. Each simulated sample consisted of a value of the statistic χ^2 calculated from the random selection of three frog models, and the sums of the corresponding score for n chicks (n = 32 chicks). The p-value was calculated as the proportion of simulated χ^2 values ≥ the actual χ^2 value (Veech 2012; Whitlock and Schluter 2015; see code availability below).

We conducted a proportional hazards survival analysis (Cox mixed-effects regression) to assess whether the risk of attack by chicks was influenced by an interaction between the presence of tadpoles on the dorsum of a frog model and the distribution of its warning coloration on the body. In this analysis, we used the latency of the chicks to attack the frog models during Phase III. As fixed effects, we included the type of frog model (i.e., full warning coloration, restricted warning coloration, without warning coloration), the quantity of tadpoles of the dorsum (i.e., no tadpoles, few tadpoles, many tadpoles), and the interaction between these two factors. As a random effect, we included the identity of each chick. Frog models completely avoided by chicks during Phase III were treated as censored data (David and Mitchel 2012; Segami Marzal et al 2017; Seymoure et al 2018).

The coefficient exponents (Exp (Coef)) from the proportional hazard in the survival analysis are estimates of relative mortality (Moore 2016; Fox and Weisberg 2018). In this analysis, we made contrasts of relative mortality between each of the different types of frog models and the baseline hazard frog model. The baseline hazard is the frog model without warning coloration with the absence of tadpoles on the dorsum in which the explanatory variables (i.e., frog model type and presence of tadpoles) are equal to zero (exp (coef) = exp (0) = 1). Therefore, values of the coefficient exponent ≥ 1 would indicate an increase in the risk of a chick attack. In contrast, coefficient exponents < 1 and ~ 0 would indicate a reduction in the risk of a chick attack. The assumptions of the survival analysis were corroborated by visual inspection of survival curves (Figure S1) and by a proportional hazards test, leaving out the "identity" effect (Moore 2016; Dessai and Patil 2019). Validation of the effect of factors included in the survival analysis was performed through factor comparison and exclusion with likelihood and Chi-square (χ^2) tests (Therneau 2015; Moore 2016). All statistical analyses were performed on the R platform (R-Development-Core-Team, 2020) with the *coxme* (Therneau, Version:2.2 2016) and *survival* (Therneau, Version:3.2-2010) packages.

Results

Phase I: learning and discrimination of the warning signal.

The seven training trials of Phase I were completed by 48 chicks. These chicks learned to discriminate between frog models with or without warning coloration (Fig. 4; Video S2). The electivity score index (ESI) of the chicks increased throughout the trials (F = 8.07, df = 4.7, p < 0.0001; n = 48 chicks). The ESI achieved by the chicks was significantly higher in the last session than in the first session (paired t = -5.736, df = 47, p < 0.001).

Phase II: learning and avoidance of the warning signal

Thirty-two (32) of the 48 chicks that performed Phase I complied with the aposematic signal avoidance criterion, i.e., they did not attack the mealworm upon the frog model with full warning coloration through three consecutive presentations (Video S2). The 32

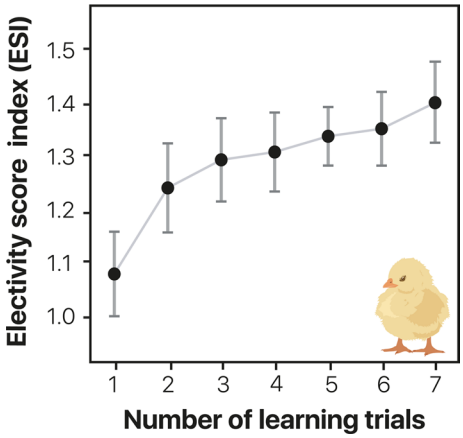

Fig. 4 Learning process (measured with the electivity score index proposed in Amézquita et al 2013) in 48 chicks *Gallus gallus domesticus* during seven learning trials of Phase I (learning and discrimination; see text for details). The electivity score index values increased when chicks attacked palatable mealworms upon frog models without warning coloration and decreased in value when the chicks attacked unpalatable mealworms (soaked in chloroquine) upon frog models with warning coloration. The black dot represents the average values, and the vertical lines represent standard errors.

chicks exhibited an average latency to attack the frog model with full warning coloration of 132 ± SD 39 s (range = 63.12–300 s). The 15 chicks that attacked the mealworm on the frog model with full warning coloration had an average latency to attack of 55 ± SD 38.39 s (range = 1.74–121 s).

Phase III: recognition of the warning signal in frog models with and without the presence of tadpoles.

The order in which chicks performed attacks on mealworms was associated with the frog model in which they were located (Friedman $\chi 2 = 67.68$, gl = 8, $p < 0.001$; Fig. 5; Video S2). Frog models without warning coloration were first attacked by chicks regardless of the presence of tadpoles on the dorsum. On the contrary, frog models with warning coloration were attacked last. Among frog models with warning coloration, those with tadpoles on the dorsum were attacked by chicks earlier than those without tadpoles. Furthermore, the order of attack by the chicks was similar between frog models without warning coloration and the frog model with restricted warning coloration having many tadpoles on the dorsum. The post hoc analysis supporting these results are shown in Fig. 5 and Table S1.

The attack behavior of the chicks toward the mealworms did not occur randomly. The Chi-square statistic (χ^2) of the observed attack scores ($\chi^2 = 41.851$) was higher than the 97.87% of the simulated Chi-square statistics under the null model, resulting in a p-value of 0.0213. Chicks attacked mealworms upon frog models without warning coloration as their first choice and avoided mealworms placed upon frog models with warning coloration. Here, we highlight that our results did not change if we assigned to the frog model with restricted warning coloration (without tadpole) a score value = 1.2 or 1.0 (Fig. 1).

The chick's latency to attack was associated to the type of frog model ($\chi^2 = 108.356$, df = 2, $p < 0.001$), the quantity of tadpoles ($\chi^2 = 13.659$, df = 2, $p = 0.001$), and the

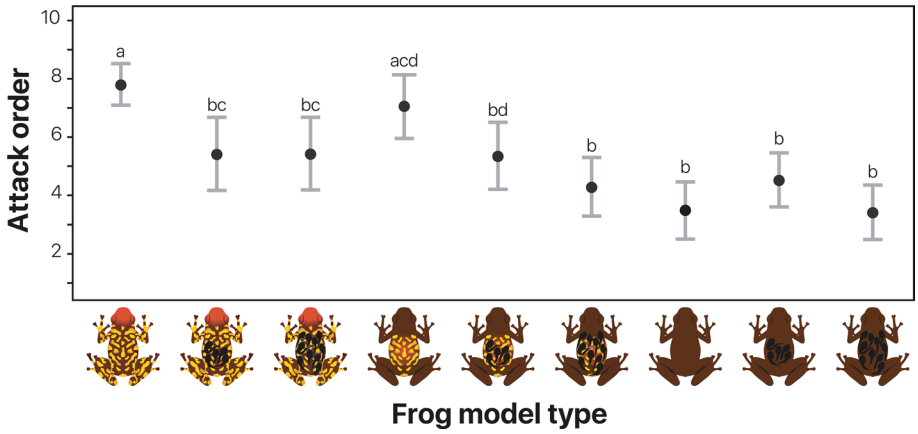

Fig. 5 Attack order by chicks to the mealworms placed upon the different frog model types (Phase III of this study). Black dots represent mean values obtained for 32 chicks; error bars represent 95% confidence intervals. Letters in each graph indicate the results of Nemenyi's post hoc pairwise comparisons test; p-values are shown in Table S1.

interaction between these two factors ($\chi^2 = 17.059$, df = 4, p = 0.001). In summary, the results of the survival analysis (Table 1) indicate that frog models with warning coloration (full and restricted) and absence of tadpoles exhibited a lower risk of attack by the chicks; frog models with full warning coloration and presence of tadpoles were the following that exhibited a lower risk of attack by the chicks; finally, frog models without warning coloration and frog models with restricted warning coloration (but with tadpoles) exhibited the highest risk of attack.

Discussion

Using domestic chicks trained to discriminate and avoid a warning signal (coloration) in printed frog models, we tested the potential predation risk that a poison frog (Dendrobatidae) might experience when they transport tadpoles on the dorsum. We obtained data supporting the three predictions put forward in this study. The results of our experiments suggest that tadpoles on the dorsum of parental individuals in aposematic species might compromise the effectivity of the warning signal display and increase the risk of being attacked by visually oriented predators. In contrast, the same results suggest that the presence of tadpoles on the dorsum in poison frogs without warning coloration would not increase the risk of attack by such potential predators.

In accordance with our first prediction (Fig. 2), frog models with warning coloration (full or restricted) and tadpoles were attacked by chicks earlier compared to frog models with warning coloration without tadpoles. This result suggests that in nature, tadpoles on the dorsum of aposematic parental frogs may interfere with warning signal recognition by visually oriented predators. Since transporting tadpoles on the dorsum in poison frogs evolved before the conspicuousness associated with aposematism (Weygoldt 1987; Carvajal-Castro et al 2021), the effect that tadpoles would have on parental predation risks could have influenced the distribution of warning coloration on the body of adult individuals. That is, conspicuous coloration would be in body parts not covered by tadpoles when

Table 1. Exponential coefficients (Exp (coef)) of the proportional hazards model (survival analysis) on chick attack latency time (Experimental Phase III). Exponent values represent the relative mortality of frog model types (with different quantities of tadpoles on the dorsum) when contrasted with the frog model without warning coloration and absence of tadpoles (Coef = 1; see methodology text for details). An Exp (coef) with values < 1 indicate a reduction in the risk of attack by the chick; conversely, values of Exp (coef) ≥ 1 indicates an increase in the probability of attack by the chick.

Frog model type	Coefficient	SE (Coef)	Exp (coef)	Z-value	p-value
	-1.665	0.284	0.189	-5.86	<0.001
	-1.525	0.271	0.217	-5.61	<0.001
	-1.034	0.277	0.355	-3.73	0.034
	-1.028	0.273	0.357	-3.76	0.031
	-0.696	0.264	0.498	-2.64	0.418
	-0.370	0.263	0.690	-1.41	0.588
	-0.333	0.261	0.716	-1.27	0.735
	0.001	0.261	1.001	0.01	3.642

Random effect	Variable	SD	Variance
ID chicks	intercept	1.076	1.159

transported on the parent's dorsum and, hence, it could still serve its warning function during this parental care behavior. According to this, in many species of Dendrobatidae, the presence of conspicuous coloration occurs towards the head and other parts of the body outside the dorsum, for example, on the limbs (e.g., *Andinobates bombetes* (Myers and Daly, 1980), *Andinobates dorisswansonae* (Rueda-Almonacid et al 2006), *Andinobates tolimensis* (Bernal-Bautista et al 2007), *Ameerega parvula* (Boulenger, 1882), *Ameerega pepperi*, Brown and Twomey, 2009; *Dendrobates tinctorius* (Cuvier, 1797); *Hyloxalus azureiventris* (Kneller and Henle, 1985), *Ranitomeya benedicta* Brown et al 2008a, b, *Ranitomeya fantastica* (Boulenger 1883). Interestingly, we do not know of species with conspicuous coloration restricted to parts of the dorsum that are potentially covered by offspring while transported. In fact, according to our hypothesis and predictions, this coloration pattern would not be favored by selection. On the other hand, in species with conspicuous coloration on the entire body of adults, the warning signal would be visible and recognized by potential predators whether the frog be transporting tadpoles (e.g., *Andinobates victimatus*

Márquez et al 2017), *Epipedobates tricolor* (Boulenger, 1899), *Oophaga pumilio* (Schmidt, 1857), *Phyllobates terribilis* Myers et al 1978).

In agreement with the second prediction (Fig. 2), the risk of attack by chicks was reduced if there was a conspicuous coloration exposed whether or not the frog models had tadpoles. This suggests that chicks based their attack decision on the conspicuous coloration, regardless of the body part of the frog model that exhibited such coloration. Other studies on birds have found that the attention at the time of deciding whether to perform an attack is focused on the coloration of the prey, irrespective of where it is distributed (Roper 1990; Gamberale-Stille and Guilford 2003; Exnerová et al 2006; Ham et al 2006; Aronsson and Gamberale-Stille 2012; Kazemi et al 2014). That is, in our study, chicks would base their attack decision on the conspicuous coloration of the frog model that is not covered by tadpoles. In several anuran lineages (e.g., *Atelopus spumarius* Cope, 1871 at Rößler et al 2019; *Bombina bombina* (Linnaeus, 1761) at Bajger 1980; *Melanophryniscus cambaraensis,* Braun and Braun, 1979 at Bordignon et al 2018) individuals exhibit conspicuous and warning coloration on body parts that would be visible only when the individual is in motion or feels threatened (Loeffler-Henry et al 2018). Something similar could occur in poison frogs traditionally classified as cryptic (e.g., *Allobates femoralis* (Boulenger, 1883), *Ameerega bilinguis* (Jungfer, 1989), *Epipedobates boulengeri* (Barbour 1905), but with small conspicuous patches in their bodies (Lötters et al 2007; Santos et al 2014; Carvajal-Castro et al 2021).

In this study, tadpoles were printed black in the frog models and, thus, could be considered analogous to black spots on conspicuously colored prey models used in several studies on aposematism and predation risk. In Qvarnström et al (2014), frog models with conspicuous colors and large black dots were attacked by chicks at higher rates than models with conspicuous colors and small or absent black dots. In this order of ideas, those results agree with ours, in which the presence of tadpoles in aposematic frog models was related to a higher attack risk by chicks (prediction 1, see Fig. 2). However, such increased attack risk on prey models with large black dots, and tadpoles printed on the dorsum, occurs in experiments using models in simple backgrounds (Aronsson and Gamberale-Stille 2008, 2009, 2012; this study). An opposite result was found under natural conditions, where clay frog models with different sizes of spots were placed on natural and complex backgrounds (Hegna et al 2011; Preißler and Pröhl 2017). Large, dark spots in colorful models could be associated with disruptive coloration, breaking the outline of the prey's body and impeding its encounter in a natural environment (Cuthill et al 2005; Stevens and Cuthill 2006; Honma et al 2015). Hence, the apparent contradiction in results between the studies mentioned here could be reflecting a distinctive effect that tadpoles (or black spots) could have on predators according to their distance to the prey (Barnett et al 2016, 2017, 2018) and movement pattern by frogs (Rojas et al 2014). For instance, at large distances between predator and prey, black tadpoles (and black spots) could hinder the detection of aposematic parental poison frogs because tadpoles on the dorsum could break the body contour of the parent in a natural environment. On the contrary, at short distances, the predator could recognize the warning coloration in body parts of the parental frog that are uncovered by tadpoles, as suggested by our results. In sum, coloration in poison frogs can be disruptive and aposematic simultaneously, and tadpoles' presence could play a role in this interplay of defensive signaling (Tullberg et al 2005; Rojas et al 2014; Barnett et al 2018). Our study emphasizing the potential effect of tadpoles on the dorsum in predation risk at short distances; further studies are necessary to test such effect at long distances.

The time spent by the parents transporting the tadpoles could be considered short for natural selection to act according to the hypotheses and predictions of this study.

However, although such time transporting the tadpoles can be short, it may be a critical stage in defining the reproductive success of the individuals. The duration of tadpole transport in poison frogs is unknown in most species (Wells 2007), including those of the *Oophaga histrionica* complex. Given that most species of poison frogs exhibit continuous reproduction and breed several times per year (Weygoldt 1987; Pröhl and Hödl 1999; Gottsberger and Gruber 2004; Lötters et al 2007; Brown et al 2008a, b; Kaefer et al 2012); the time a parental is transporting tadpoles on the dorsum might be longer than expected. Another aspect that could influence predation risk during tadpole transport is the distance between the egg deposition site and water bodies where tadpoles will develop. Parents can move long distances while transporting tadpoles (Pašukonis et al 2019) which increases predation risks and the possibility that selection promotes color traits such as those predicted in this study. This increase in predation risk with transporting distance can be especially high in aposematic species that climb on trees looking for phytotelmata (Carvajal-Castro et al 2021). Overall, predation events during tadpole transport have been recorded in aposematic species such as *Amereega trivittata* (Spix, 1824) (Ramos and Caicedo-Moncada 2019; Pašukonis and Loretto 2020) and non-aposematic species such as *Allobates femoralis* (Beck et al 2017). Nevertheless, more data is necessary to make a definitive conclusion, especially because the influence of tadpole presence on color evolution may vary among clades with different color patterns, predator pressures, and parental care modes.

Several factors may have influenced the results obtained during the experiments of this study. First, neophobia is the innate fear that many organisms may present to a certain perceived novel trait (Schuler and Hesse 1985; Mastrota and Mench 1995; Lindström et al 1999). However, all of the 84 chicks attacked the frog model with full warning coloration on the first encounter (Phase I); therefore, the subsequent decrease in attack rate toward other frog models indicates a learning process and rules out neophobia (Roper and Wistow 1986). Second, patterns in the warning signal (e.g., reticulations) could alter the attack rate by inexperienced chicks (Halpin et al 2020). Differences in attack behavior during Phase III by chicks can be assigned to the presence of tadpoles on the dorsum in frog models and not due to internal features in the pattern of the warning signal because chicks were trained using the same frog models (Phases I and II). Third, environmental characteristics may generate variability in the perception and detectability of warning coloration by predators; for example, changes in illumination and spatial heterogeneity of the habitat (Dreher et al 2015; Barnett et al 2021; Nokelainen et al 2022). Future research could use clay models under natural conditions and salivary DNA traces (DNA barcoding or metabarcoding) of natural predators to assess the relative importance of habitat characteristics and type of predator shaping the parental care-aposematism relationship we are testing (Bateman et al 2017; Casas-Cardona et al 2018; Rößler et al 2020).

In poison frogs, transporting tadpoles on the dorsum implicates energy costs, time investment, reduction of mating opportunities, and risk of losing a territory (Beck et al 2017; Ringler et al 2013; Wells 2007; Pašukonis et al 2019). This study provides evidence of an additional cost of parental care in aposematic poison frogs: an increase in the mortality risk due to the interference that tadpoles may create in recognition of the warning signal (coloration) by visually oriented predators. However, the function of the warning signal can be maintained if it is on parts of the parental body that are not covered by tadpoles during transport to bodies of water. Parental care is a trait present in a high diversity of organisms, many of them aposematic (Trivers 1972; Clutton-Brock 1991; Royle et al 2012). Therefore, the results of this study contribute to the understanding of the cost associated

with parental care, and factors and processes that may underlie interactions between predation risk, parental care, and the diversity of warning signals in nature.

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Authors' contributions All authors contributed to the study conception and design. J.D.C.C., S.C.C, and F.V.S developed the research concept. Experiments, data collection and analysis were performed by M.P.T.G. All authors planned the research, interpreted the data, read, edited, and approved the final manuscript.

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Availability of data and material Data supporting the findings reported in this study are available at the FigShare data repository https://figshare.com/s/d28c09abc114b61c18a9

Code availability Script used in this study is available at the FigShare data repository https://figshare.com/s/0855586b4a985625214e

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

Ethical Statement The authors followed ethical procedures of the Guidelines for the treatment of animals in behavioral research and teaching by the Association for the Study of Animal Behaviour during all the observations and experiments performed in this study. Additionally, this study was made under research permit # 01789 (October 11, 2018) of the government of Colombia.

Consent to participate All authors consent to participate.

Consent for publication All authors approved the manuscript.

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